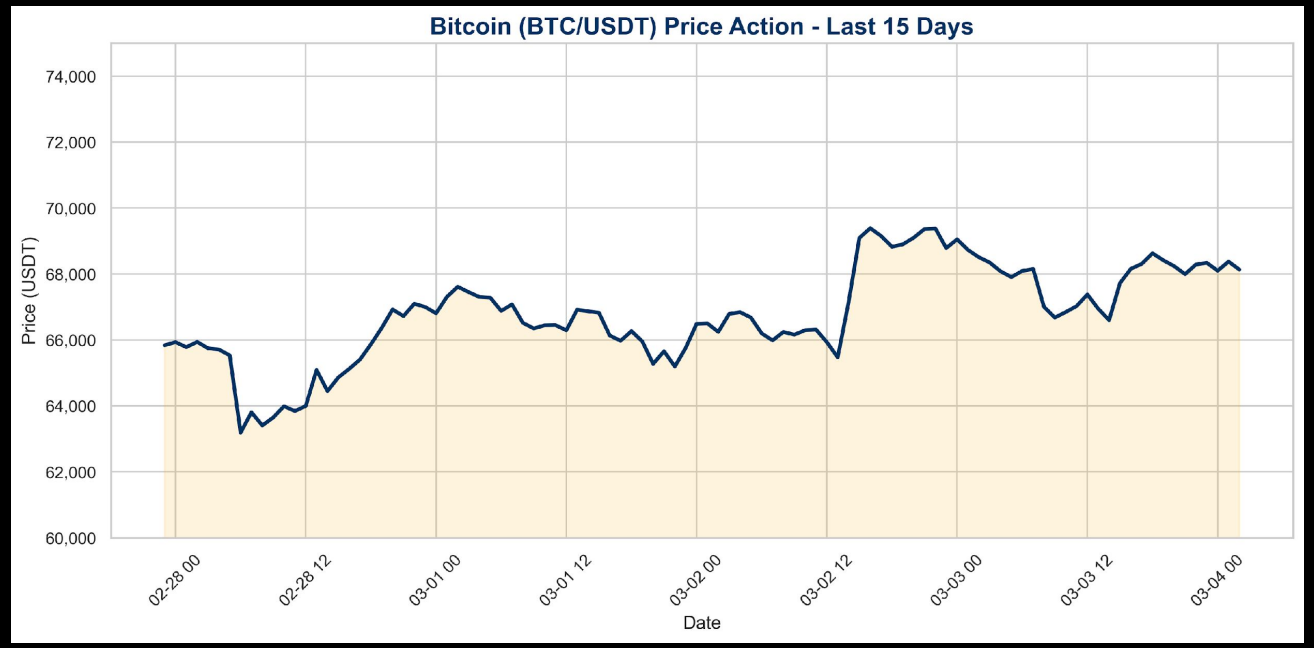
DEEP REINFORCEMENT LEARNING & PATH SIGNATURES FOR CRYPTO TRADING

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THE CHALLENGE: OVERFITTING IN CHAOTIC MARKETS

- Financial data is incredibly noisy and non-stationary.
- Standard RL agents memorize the noise (In-Sample Overfitting).
- Result: Complete failure on unseen data (Out-of-Sample Degradation).



OUR SOLUTION: RL + ROUGH PATH THEORY

Path Signatures

- Filters noise
- Extracts Sharpe
- Scale-invariant

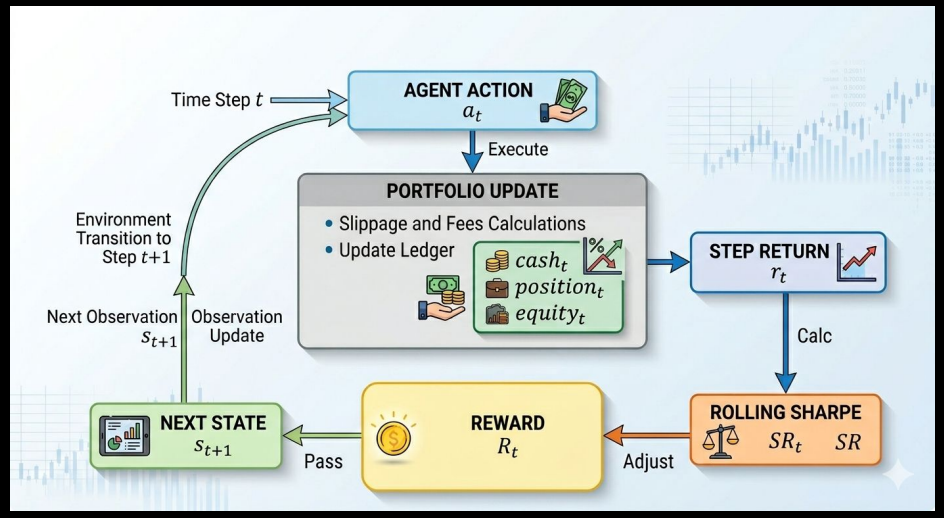
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Deep RL

- Will learn from clean data $\rightarrow$ less overfitting
- Maximize sharpe ratio

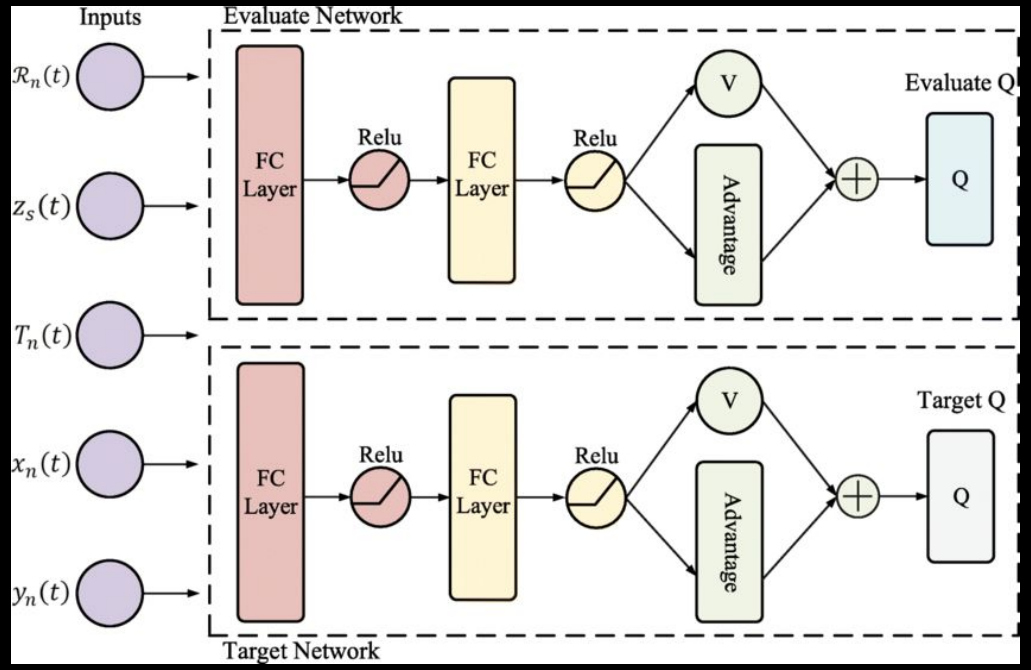
Methodology: Capital-Aware Environment

- Replaced (+1,0,−1) action scheme to more actions and a continuous trading environment.
- Built a capital-aware Gymnasium environment with explicit cash/position/equity accounting.
- Added market frictions and controls: 0.1% fee, slippage, fractional actions, and strict exposure/risk constraints.



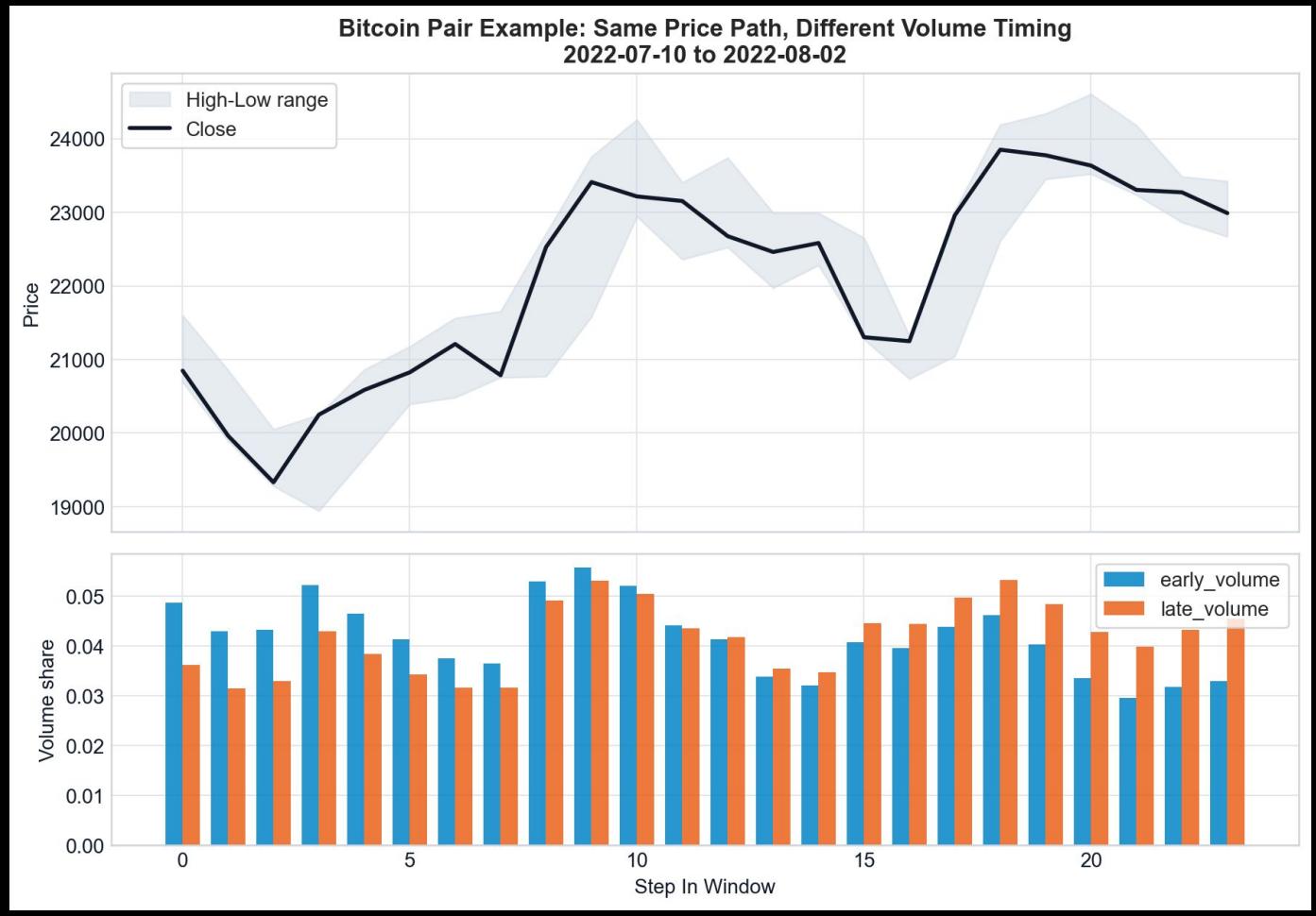
Methodology: D3QN

- Primary Architecture: Deep Double Dueling DQN (D3QN).
- Key Upgrades: Prioritized Experience Replay (PER) & n-step returns to efficiently process complex signature states.
- Robust Benchmarking: Proximal Policy Optimization (PPO) & Soft Actor-Critic (SAC) implemented for continuous action space comparison.



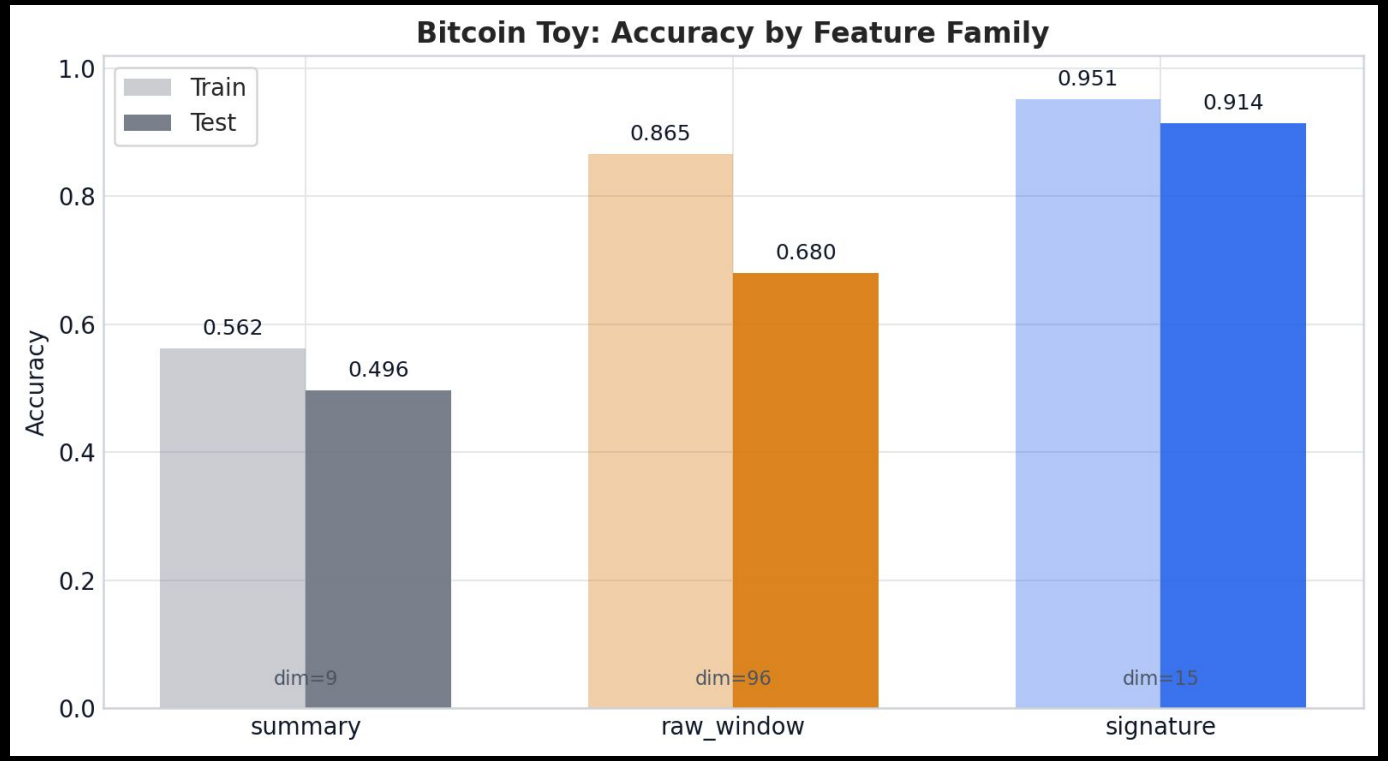
Experiments: Volume-Timing Representation

- Bitcoin OHLCV windows from the project dataset.
- Close / High / Low are held fixed across the pair.
- Only within-window volume timing changes: early volume vs late volume.



Experiments: Volume-Timing Representation

Feature	Dim	Train Acc.	Test Acc
summary	9	0.5625	0.4961
raw window	96	0.8652	0.6797
signature	15	0.9512	0.9141



Preliminary Results

- Strict temporal split: Train/Validation (IS) 2014-2020 | Test (OOS) 2021-2024.
- PPO Baseline: Currently exhibits the strongest generalization (OOS Sharpe: 1.1079).
- D3QN Performance: Excellent for ambitious trading: OOS Return of 85.86%, OOS Sharpe Ratio of 0.8905.

Method	OOS Sharp Ratio	OOS Avg. Return
PPO	1.1079	66.35
D3QN (3-action)	0.8650	68.60
D3QN (6-action)	0.8905	85.86

Ablation Results Across Feature Variants

Config	5-seed Sharpe	5-seed Return %
Baseline	**0.8905**	**85.86**
D3_return_vol5 (drop log_price)	**0.8728**	**95.38**
F1_hlrange_rv10 (add hlrange, widen vol)	**0.9072**	**84.72**
MLP_D3_return_vol5 ***(MLP encoder)***	**0.8073**	**77.99**
MLP_R5_volprof (swap to norm. volume)	**0.8059**	**100.12**

Key observations

- Baseline remains strong.
- D3_return_vol5 raises return but slightly lowers Sharpe.
- F1_hlrange_rv10 has the best Sharpe.
- MLP_D3_return_vol5 underperforms.
- MLP_R5_volprof has the highest return but weaker Sharpe.

Conclusion

Transition into multi asset portfolio

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